

NETWORKED CONTROL

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NAME/FAMILY NAME..... SOLUTIONS.....
PERSONAL CODE.....
SIGNATURE.....
DATE..... 12/07/2022.....

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

$$1) \begin{cases} \dot{x} = 2u \\ y = 2x \end{cases}$$

$$A) u = ky = 2kx$$

$$\Rightarrow \dot{x} = 4kx \quad \text{AS. STABLE IFF}$$

$$k < 0$$

$$B) h=1$$

$$F=1, G=2h=2, H=2$$

$$\Rightarrow x_{k+1} = x_k + 2u_k$$

$$y_k = 2x_k$$

$$u_k = ky_k = 2kx_k \Rightarrow x_{k+1} = x_k + 4kx_k \\ = (1+4k)x_k$$

$$\text{AS. STABILITY IFF } |1+4k| < 1, \text{ i.e.,}$$

$$-1 < 1+4k < 1 \quad (\Leftrightarrow) \quad \begin{matrix} 4k > -2 \\ 4k < 0 \end{matrix} \quad (\Leftrightarrow) \quad -\frac{1}{2} < k < 0$$

$$C) h=1s, k=-\frac{1}{4}$$

$$a) \text{ Define } z_k = \begin{bmatrix} x_k \\ y_{k-1} \end{bmatrix} \Rightarrow z_{k+1} = \mathbb{I}(h, z) z_k$$

$$\mathbb{I}(h, z) = \begin{bmatrix} 1+4k(h-z) & 2kz \\ 2 & 0 \end{bmatrix}$$

$$\text{IF } h=1s, z=0s$$

$$\Rightarrow \mathbb{I}(h, z) = \begin{bmatrix} 1-1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \Rightarrow \text{AS. STABILITY}$$

b) $h=1 \rightarrow, z=0.2$

$$\Phi(h, z) = \begin{bmatrix} 1 - (1-0.2) & -\frac{1}{2}(0.2) \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.1 \\ 2 & 0 \end{bmatrix}$$

$$p(\lambda) = \det(\lambda I - \Phi(h, z)) = \det \begin{bmatrix} \lambda - 0.2 & 0.1 \\ -2 & \lambda \end{bmatrix}$$

$$= \lambda^2 - 0.2\lambda + 0.2$$

$b = 0.2 \quad |b| < 1 \quad \xrightarrow{OK}$

$a = -0.2 \quad |a| < 1.2 \quad \xrightarrow{OK}$

$\Rightarrow$ AS. STABLE

c) $\mathbb{I}_{k+1} = \tilde{\Phi}_{0k}(h, z) \mathbb{I}_k$

$$\tilde{\Phi}_1(h, z) = \Phi(h, z) = \begin{bmatrix} 0.2 & -0.1 \\ 2 & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0(h, z) = \begin{bmatrix} 1 & 2K \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix}$$

For $k = 1+3i, i \in \mathbb{N}$:

$$\mathbb{I}_{k+3} = \underbrace{\tilde{\Phi}_0 \tilde{\Phi}_0 \tilde{\Phi}_1}_{\tilde{\Phi}_{tot}} \mathbb{I}_k = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.2 & -0.1 \\ 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.2 & -0.1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -1.8 & -0.1 \\ 2 & 0 \end{bmatrix}$$

$$p(\lambda) = \det(\lambda I - \tilde{\Phi}_{tot}) = \det \begin{bmatrix} \lambda + 1.8 & 0.1 \\ -2 & \lambda \end{bmatrix}$$

$$= \lambda^2 + 1.8\lambda + 0.2$$

$b = 0.2 \quad |b| < 1 \quad r$

$a = 1.8 \quad |a| \not< 1+b \quad x$

$\Rightarrow$ NOT AS. STABLE!

INDEED $\lambda = -0.9 \pm \sqrt{0.45 - 0.2}$
 $= -0.9 \pm \sqrt{0.25} = -0.9 \pm 0.5 \begin{cases} -1.4 & \times \\ -0.4 \end{cases}$

$$|\lambda| = 0.45 + 0.35$$

2) SEE THEORY

3) A, B SEE THEORY

c) $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$

$$p(\lambda) = \det \begin{bmatrix} \lambda - 2 & -1 \\ -3 & \lambda - 2 \end{bmatrix}$$

$$\begin{aligned} &= \lambda^2 - 4\lambda + 4 - 3 \\ &= \lambda^2 - 4\lambda + 1 \end{aligned}$$

$$p(\lambda) = 0 \Rightarrow \lambda_{1,2} = 2 \pm \sqrt{4 - 1} = 2 \pm \sqrt{3} \begin{cases} 3.732 \\ 0.2679 \end{cases}$$

$$\frac{N_B}{h} \geq (2 + \sqrt{3} + 2 - \sqrt{3}) \log_2 e = 4 \log_2 e \approx 5.76$$

$$N_B \geq 5.76 \cdot h \approx 1.15$$

The maximum # of bits for
 boundability is 2.

